

MOHAMED ASAAD

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OBJECTIVE

Electrical Engineer with expertise in renewable energy, power systems, and electrical design. Passionate about sustainable solutions, power optimization, and innovative engineering approaches. Proven experience in transformer design, power quality assessment, and industrial energy efficiency. Seeking an opportunity to apply my skills in electrical engineering and energy solutions.

EXPERIENCE

Research and Development Engineer

TECTRAFO | Giza, Egypt

April 2025 – Present

- Led R&D initiatives to enhance transformer design methodologies and performance.
- Developed a custom transformer design software using Python and Excel for process automation.
- Implemented improvements that reduced design errors and improved accuracy.

Electrical Design Engineer

TECTRAFO | Giza, Egypt

Jun 2024 – April 2025

- Designed and optimized transformers in accordance with IEC and IEEE standards.
- Created detailed technical documentation, drawings, and specifications.

EDUCATION

M.Sc. in Electrical Power & Machines Engineering

Ain Shams University | Cairo, Egypt

2023 – Present

Energy Management for Smart Buildings with AI – Ongoing Research Thesis | GPA: 3.2/4.0

B.Sc. in Electrical Power & Machines Engineering

Ain Shams University | Cairo, Egypt

2017 – 2022

GPA: 2.3/4.0

CERTIFICATIONS & TRAINING

- Renewable Energy and Green Building Entrepreneurship | Duke University (2024)
- Electrical Production Engineer | Arab International Optical Company (2023-2024)
- Scientific Writing for MSc Students | Ain Shams University (2024)
- Test of English Language Proficiency (TELP) | Ain Shams University (2024)
- Fundamentals of Digital Marketing | Google Digital Garage (2023)
- Employability Skills Training | Ain Shams University Career Center (2022)
- UP SKILL Program (Modules 1-4) | Ain Shams University Career Center (2024)
- Electrical Design Diploma | Dar Al-Kahrbaa Training Center (2022)
- Assistant Engineer | Cairo Roads and Construction Company (2018)

PROJECTS

Transformer Design Software

TEC TRAFO | Giza, Egypt

Jun 2024 – Present

- Developed transformer design software to automate and optimize transformer design processes.

Integrated Power Quality Assessment and PV Solution for Industrial Facilities

Danone Factory | Obour, Egypt

Aug 2021 – Jun 2022

- Led a solar PV design and power quality study for industrial loads.
- Delivered performance reports and design proposals for energy savings.

SKILLS

Electrical Design & Power Systems: Comprehensive electrical infrastructure design, MV & LV systems, AutoCAD, ETAP, PSCAD, MATLAB

Renewable Energy: Solar PV design (PVsyst), grid-connected and off-grid solutions

Smart Solutions & Data Analytics: Energy management, data-driven optimization, forecasting models (XGBoost, LSTM), machine learning frameworks (TensorFlow, Keras)

Low Voltage Systems: Fire alarm, CCTV, access control, data, and telephone systems

Project Management: Technical documentation, project lifecycle management, risk assessment and mitigation

Compliance & Standards: Adherence to IEC, IEEE, and local regulations

Enterprise Resource Planning (ERP): SAP system

Programming Proficiency: Python

English: Proficient (TELP Score: 550)